

rooms. Examples include SCARA and six-axis robots from Epson, FANUC, Yamaha, and ABB.

Automotive and electric vehicles.

Compared with conventional automotive robots, earlier versions were used primarily for welding and painting. In electric vehicle production, robots are increasingly performing new tasks. They are assembling battery modules, applying adhesive, and handling power electronics. Articulated robots such as ABB's IRB 2600, KUKA's KR Cybertech, Yaskawa's GP series, and Epson's VT-Series are widely used for these applications.

Logistics and warehousing. In the logistics industry, robots operate on conveyor belts, and autonomous mobile robots are used to sort items, stack boxes, and prepare orders more quickly. In such applications, system-level expertise and availability are more important than robot performance.

Why industry-ready engineering has become a strategic issue

While robotics technology continues to advance rapidly, one constraint persists: the shortage of engineers capable of implementing and maintaining increasingly complex systems. Modern robotics extends well beyond basic motion programming. It requires knowledge across mechanical engineering, electrical engineering, control software, AI, machine vision, and safety compliance. Few engineering programmes address this full spectrum.

To bridge the gap, the industry is adopting hands-on education models that bring the factory floor into the classroom. Students work directly with industrial robots, SCARA arms, and six-axis systems, learning not only theory but also system integration. Epson, for example, has established robotics laboratories at Ramaiah Institute of Technology in Bengaluru and plans to establish a similar facility

at Vellore Institute of Technology in Chennai. Students from multiple institutes have spent several weeks training on production-grade equipment and completing internships at industrial solution centres. The emphasis is on practical learning rather than purely theoretical instruction.

These labs typically operate on a shared-investment model. Industry provides equipment, technical staff, and training support, while institutions supply space, faculty, and curriculum integration. Students are not charged to use the facilities. The goal is workforce readiness rather than certification revenue. Preparing graduates for real-world deployment is treated as a shared responsibility between companies and colleges. Building human capability is therefore viewed as equally important as advancing the technology itself.

What needs to happen next: Scaling the model

As robotics adoption accelerates, education initiatives must expand in scale and reach. Programmes should extend beyond tier-one institutions to include state universities, polytechnics, and regional engineering colleges so that a broader base of students gains exposure to robotics and automation.

Exposure alone, however, is insufficient. Faculty upskilling is essential, and robotics, AI, and automation concepts must be integrated into core engineering curricula rather than treated as add-ons. Industry-standard certification pathways can help students demonstrate competencies aligned with factory requirements. Long-term partnerships between industry and academia are also critical to ensure that educational content evolves alongside technological change.

For automation providers, the objective goes beyond maintaining technical leadership. Sustained

engagement with education helps ensure that engineers are prepared for the next generation of smart factories. Workforce development is not a one-time effort but a continuous process, much like robotics innovation itself.

The Indian context: Make in India and the robotics imperative

The emphasis on local manufacturing through Make in India and PLI schemes has made robotics a necessity rather than a luxury. As global manufacturers expand domestic production, the availability of qualified automation engineers becomes a key competitive factor. Robotics education now sits at the intersection of industrial policy, talent development, and technological self-reliance.

Industry-academia collaborations, implemented at scale and sustained over time, can align engineering talent with the requirements of next-generation manufacturing. The future of robotics will be shaped not only by smarter machines but also by how effectively people are prepared to work with them. Edge AI, vision-guided automation, collaborative robots, and flexible production cells are already reshaping factories worldwide. Without engineers trained across hardware, software, and intelligence, the potential of these technologies cannot be fully realised.

Recent initiatives underscore that producing industry-ready engineers is no longer optional but essential. The pace at which India adopts advanced robotics over the coming decade will depend as much on workforce capability as on machine capability. Smarter factories will require smarter engineers. **EFY**

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